IPW3 : NACA0012 Code-to-Code Comparisons

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September 21, 2026

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Geometry and Aerodynamic Conditions

NASA Icing Research Tunnel

Airfoil	NACA0012
Chord, c	$0.5334\ \mathrm{m}$
Angle of attack	4°

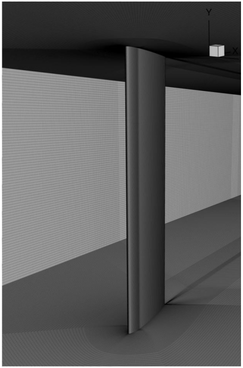
Aerodynamic Conditions

M_∞	0.3114
U_∞	$102.15\ \mathrm{m/s}$
T_∞	$267.75\ \mathrm{K}$
ρ_∞	$1.1884\ \mathrm{kg/m^3}$
Re_c	3.8441×10^6

Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$

Icing Conditions



Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	**AE3933**
T_0 [°C]	-5.6	-2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m^3]	0.55	0.55
MVD [μm]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	**101.0**	**85.9**

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g). This difference will be revisited when assessing the predicted thermodynamic mass balance and ice accretion.

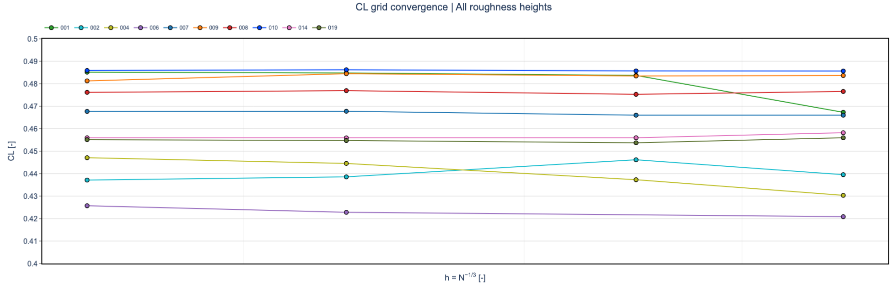
Droplet Distribution

- Single-bin to 15-bin distributions
- Total LWC preserved between distributions
- Evaluated across all four grid levels

Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad f, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$

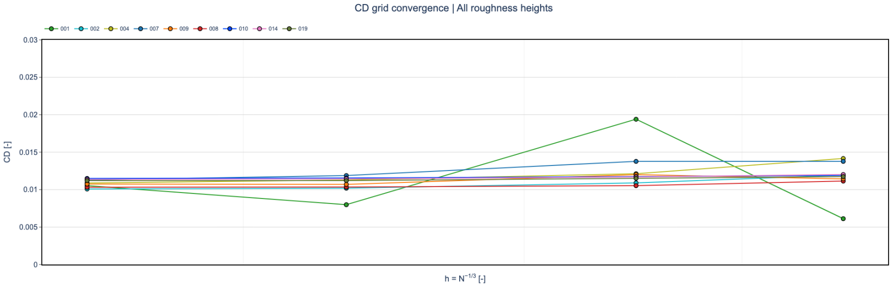
NACA0012 – Lift Coefficient Grid Convergence



Assessment

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit limited sensitivity to spatial grid refinement.
- The largest coarse-grid deviations are approximately 4% relative to the corresponding L1 predictions.

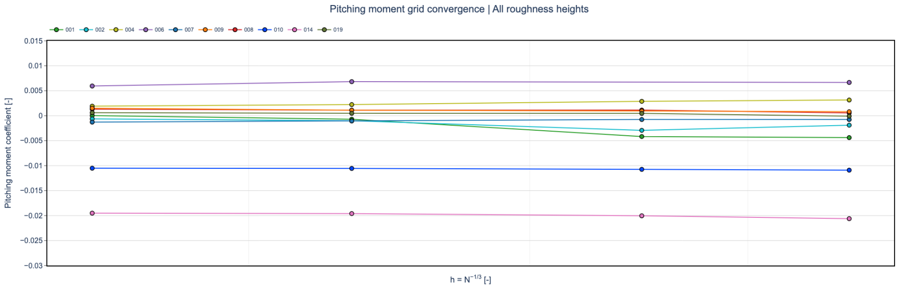
NACA0012 – Drag Coefficient Grid Convergence



Assessment

- At L1, the median drag coefficient is $C_D = 0.01088$, with an IQR of 7.57×10^{-4} (7.0%).
- C_D exhibits greater sensitivity to spatial grid refinement than C_L.
- Non-monotonic grid-convergence behavior is observed for selected submissions.

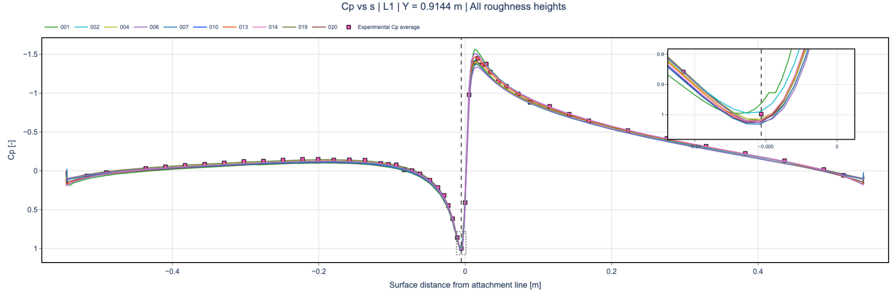
NACA0012 – Pitching Moment Grid Convergence



Assessment

- At L1, the median pitching-moment coefficient is $C_M \approx 0$, with an IQR of 2.60×10^{-3}.
- Most individual submissions exhibit limited sensitivity to spatial grid refinement.
- Code-to-code offsets are larger than the grid-refinement effect for most submissions.
- Relative percentage differences are not meaningful for C_M because the reference values are close to zero.

NACA0012 – Surface Pressure Distribution (L1)



Assessment

- The submitted C_p distributions are closely clustered and show good agreement with the experimental measurements over most of the airfoil.
- The largest code-to-code differences are concentrated near the leading-edge suction peak and stagnation region.
- The overall pressure distribution is consistent across submissions despite differences in prescribed surface roughness.

NACA0012 – Aerodynamic Assessment Summary

Grid Convergence

- C_L: generally weak grid sensitivity, with most submissions remaining close to their L1 prediction.
- C_D: greater grid sensitivity, with non-monotonic behavior observed for selected submissions.
- C_M: generally limited within-code grid sensitivity, while code-to-code offsets remain comparatively large.

Code-to-Code Variability at L1

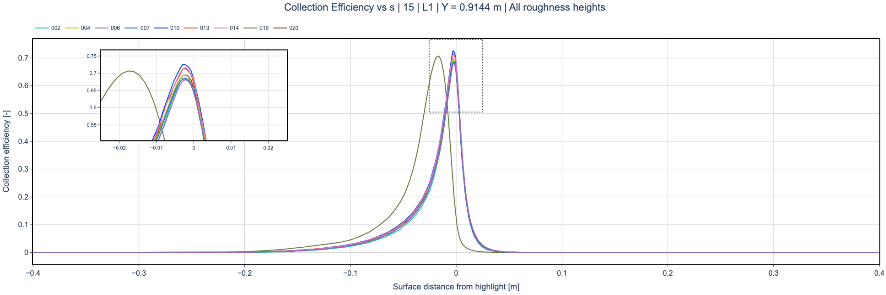
	Median	IQR	Relative IQR
C_L	0.4677	0.0262	5.6%
C_D	0.01088	7.57×10^{-4}	7.0%
C_M	≈ 0	2.60×10^{-3}	–

Relative IQR is not reported for C_M because normalization by a value close to zero is ill-conditioned.

Aerodynamic Takeaway

The C_p distributions are closely clustered and generally agree well with the experimental measurements. Grid refinement produces relatively small changes in C_L and C_M for most submissions, while C_D is more grid-sensitive. Residual code-to-code differences therefore cannot be attributed to spatial resolution alone.

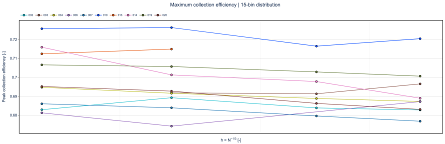
NACA0012 – Collection Efficiency

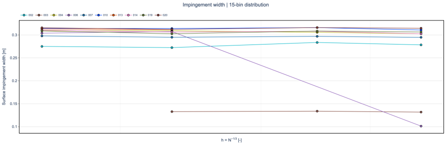


Assessment

- Most submissions predict a similar collection-efficiency distribution around the leading edge.
- Peak collection efficiencies are closely clustered, while differences remain in the detailed profile and impingement extent.
- Differences in prescribed surface roughness should be considered when interpreting the remaining code-to-code variability.

NACA0012 – Collection Efficiency Grid Convergence

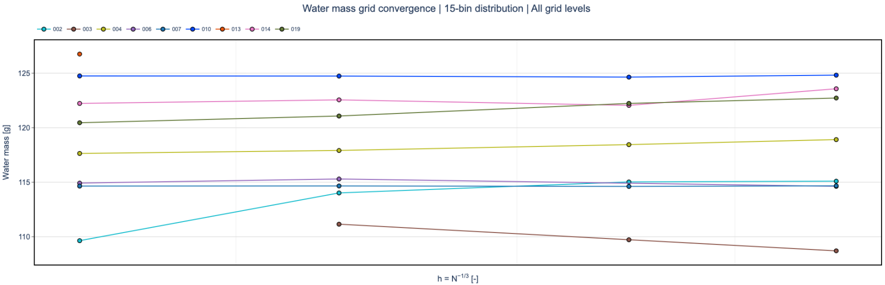




Assessment

- At L1, the median peak collection efficiency is $\beta_{\text{max}} = 0.695$ with an IQR of 0.0209 (3.0%).
- The L1 median impingement width is $0.309\ \mathrm{m}$ with an IQR of $8.93 \times 10^{-3}\ \mathrm{m}$ (2.9%).
- Both quantities exhibit limited grid sensitivity for most submissions, indicating comparatively stable local impingement predictions.

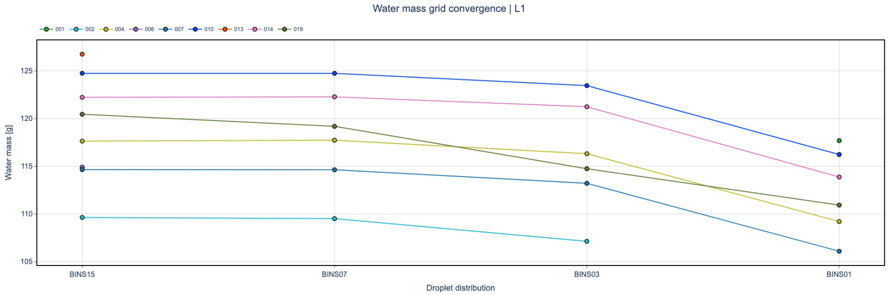
NACA0012 – Water Mass Grid Convergence



Assessment

- For the 15-bin distribution, the L1 median water mass is 119.05 g with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity, with changes generally within approximately 5% of their L1 prediction.
- Code-to-code differences are generally larger than the grid-refinement effect.

NACA0012 – Droplet Distribution Sensitivity



Assessment

- The 15-bin and 7-bin distributions produce very similar impinged water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest change for most submissions.
- The single-bin approximation reduces the collected water mass by approximately 5–8% relative to the 15-bin distribution for most participants.

NACA0012 – Droplet Impingement Summary

Spatial Grid Convergence

- Weak grid sensitivity is observed for most submissions across both local and integrated impingement quantities.
- At L1, the code-to-code spread is 3.0% for β_{max}, 2.9% for impingement width, and 5.37% for integrated water mass.
- Code-to-code variability generally exceeds the effect of spatial grid refinement.

Droplet-Size Distribution

- The 7- and 15-bin distributions provide very similar integrated water-mass predictions.
- The single-bin approximation predicts approximately 5–8% less collected water than the 15-bin distribution for most submissions.
- Diameter-resolved results show greater scatter for the smallest and largest droplets, while bins near the MVD are more consistent.

Icing-Condition Sensitivity

- Little difference in total water impingement is observed between the two NACA0012 icing conditions.

Takeaway

Spatial grid refinement has a limited influence on the predicted impingement, whereas droplet-size discretization introduces a more systematic effect. The 7-bin distribution appears sufficient to reproduce the integrated 15-bin result.

NACA0012 – Convective Heat Transfer Evaluation

Integrated Convective Heat Transfer

The convective heat transfer is evaluated along the surface slice as

$$\boxed{Q'_c = \int HTC(s)\,[T_s(s) - T_{\mathrm{rec}}(s)]\;ds} \qquad [Q'_c] = \mathrm{W/m}.$$

The submitted HTC, surface temperature T_s, and pressure coefficient C_p are used directly.

Recovery Temperature Reconstruction

Since T_{rec} is not directly available from the submissions, it is reconstructed from the submitted C_p distribution:

$$\boxed{C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{\mathrm{rec}}(s)}$$

Assuming $p_s \simeq p_e$ and an isentropic external flow,

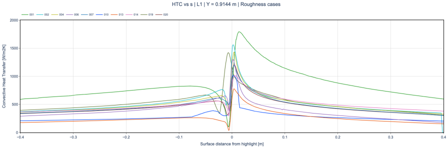
$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_p.$$

Recovery Model

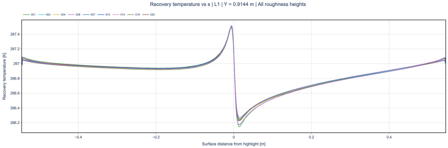
For a turbulent boundary layer,

$$T_{\mathrm{rec}} = T_e \left[1 + r\frac{\gamma - 1}{2} M_e^2\right], \qquad r = Pr^{1/3} \approx 0.9.$$

NACA0012 – Convective Heat-Transfer Distributions (L1)



Convective heat-transfer coefficient

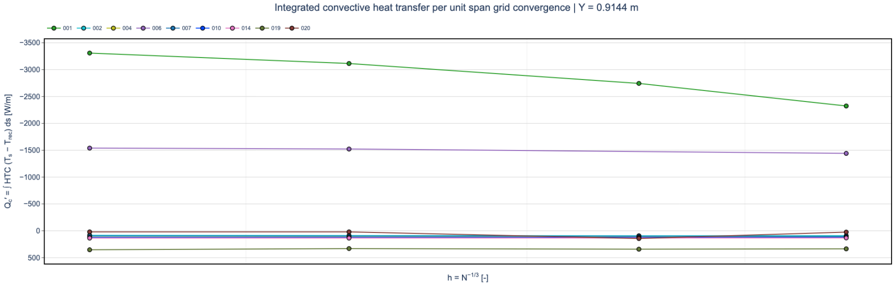


Reconstructed recovery temperature

Assessment

- On the L1 grid, reconstructed recovery-temperature distributions are closely clustered across submissions.
- Considerably larger code-to-code differences are observed in HTC, particularly near the leading edge.
- The L1 comparison indicates that the variability in the convective thermal boundary condition is primarily associated with HTC rather than T_{rec}.
- Differences in prescribed surface roughness should be considered when interpreting the remaining HTC variability.

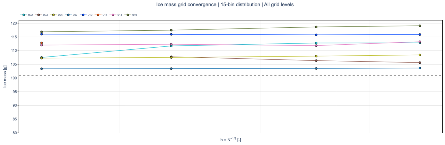
NACA0012 – Convective Heat Transfer Grid Convergence



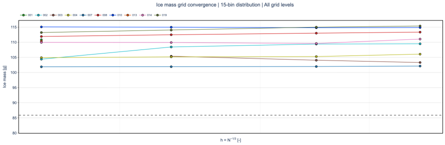
Assessment

- For most submissions, the integrated convective heat transfer exhibits relatively limited variation with spatial grid refinement.
- Code-to-code variability is substantially larger than the grid-refinement effect.
- The observed spread is primarily associated with differences in the submitted *HTC* distributions rather than the reconstructed recovery temperature.
- A small number of submissions exhibit markedly different integrated heat-transfer behavior and should be distinguished from the main cluster.

NACA0012 – Ice Mass Grid Convergence



AE3932

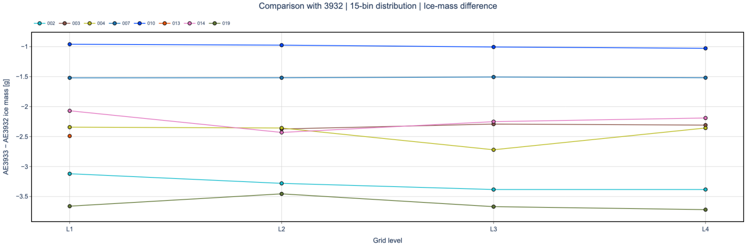


AE3933

Assessment

- Most submissions exhibit limited ice-mass sensitivity to spatial grid refinement for the 15-bin distribution.
- An isolated submission exhibits substantially stronger grid sensitivity than the main participant cluster.
- Code-to-code differences remain visible on the finest grid and cannot be attributed solely to spatial resolution.
- A systematic difference between AE3932 and AE3933 begins to emerge in the predicted ice mass.

NACA0012 – Ice Mass Difference Between Conditions



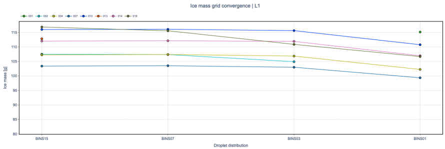
Assessment

- The matched-participant comparison directly isolates the ice-mass difference between AE3933 and AE3932.
- For most submissions, AE3933 predicts less ice mass than AE3932.
- The case-to-case difference remains comparatively consistent across the grid hierarchy for most submissions.
- This is the first clear systematic distinction between the two conditions after broadly comparable aerodynamic, impingement, and convective heat-transfer results.

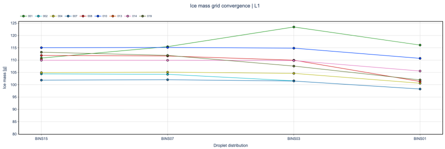
Key Observation

Despite comparable aerodynamic, impingement, and convective heat-transfer predictions, the two conditions begin to separate in the resulting ice mass. However, the approximately 15% difference observed experimentally between the two conditions is not reproduced by the numerical predictions, which generally show a substantially smaller separation.

NACA0012 – Ice Mass Sensitivity to Droplet Distribution



AE3932

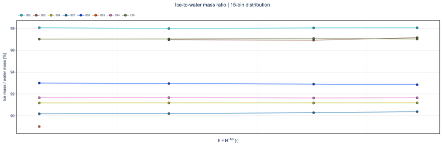


AE3933

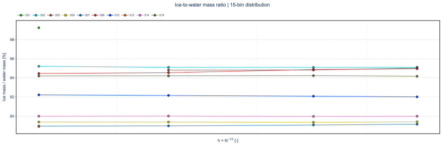
Assessment

- The 15- and 7-bin distributions generally produce similar ice masses.
- Greater sensitivity appears as the droplet distribution is reduced to 3 and particularly 1 bin.
- For most submissions, reducing the droplet distribution modifies the ice mass more noticeably than the 15-to-7-bin reduction.
- The response is participant-dependent, indicating that droplet-size discretization influences the resulting icing response in addition to total water impingement.

NACA0012 – Ice-to-Water Mass Ratio



AE3932



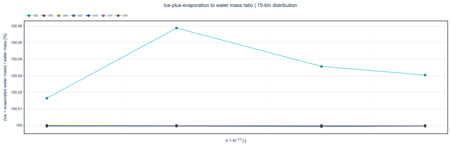
AE3933

$$\frac{m_{\text{ice}}}{m_{\text{water}}}$$

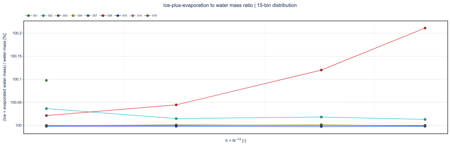
Assessment

- Normalizing by the impinged water mass separates the icing response from differences in total water collection.
- Since the two conditions exhibit comparable water impingement, differences in this ratio directly highlight differences in the resulting ice accretion.
- The comparison therefore provides a clearer measure of the transition from impingement to the thermodynamic icing response.

NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932



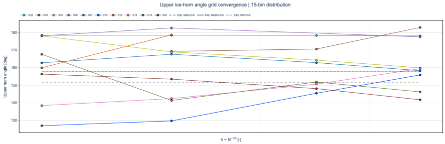
AE3933

$$\frac{m_{\text{ice}} + m_{\text{evap}}}{m_{\text{water}}}$$

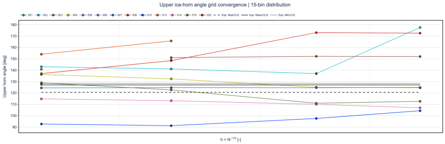
Assessment

- Including evaporated water provides additional information on the partition of the impinged water mass.
- Comparison with $m_{\text{ice}}/m_{\text{water}}$ identifies the contribution of evaporation to the difference between the two icing conditions.
- The remaining fraction represents water leaving the local ice-accretion balance through the other modeled mass-transfer mechanisms.

NACA0012 – Upper Horn Angle Grid Convergence



AE3932

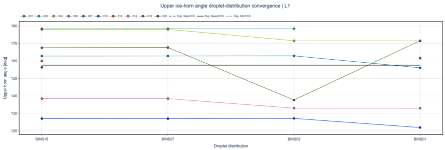


AE3933

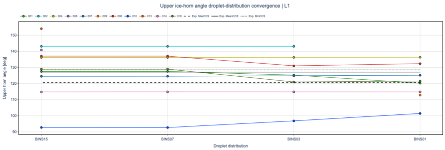
Assessment

- The upper horn angle provides a geometric grid-convergence metric complementary to the integrated ice mass.
- Differences in horn orientation quantify changes in where the accreted ice is distributed rather than only how much ice is formed.
- Comparison between AE3932 and AE3933 indicates whether the change in icing response also produces a systematic change in horn orientation.

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution



AE3932

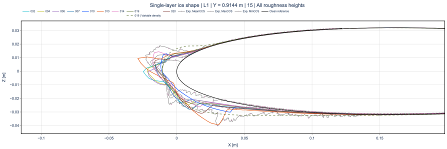


AE3933

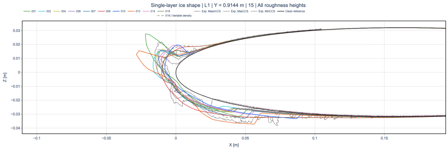
Assessment

- The L1 comparison evaluates the sensitivity of the upper horn orientation to droplet-size discretization.
- This provides a geometric counterpart to the ice-mass droplet-distribution sensitivity.
- Changes in total ice mass and changes in horn orientation should therefore be considered together when assessing droplet-distribution convergence.

NACA0012 – Ice Shapes (L1, 15 Bins)



AE3932



AE3933

Assessment

- The L1 15-bin ice shapes provide the final geometric comparison of the two icing conditions.
- The comparison reveals how differences in ice mass and mass partition translate into horn orientation, local thickness, and overall accretion extent.
- Experimental ice shapes provide the validation reference for the predicted spatial accretion pattern.

NACA0012 – Ice Accretion Summary

Ice Mass

- Most submissions exhibit limited 15-bin ice-mass sensitivity to spatial grid refinement, with isolated stronger grid dependence.
- In contrast to the preceding quantities, a systematic difference begins to emerge between AE3932 and AE3933 in the resulting ice mass.

Mass Partition

- The ice-to-water mass ratio separates the icing response from the comparable water impingement between the two conditions.
- Including evaporation further identifies how the impinged water is partitioned between ice formation and other mass-transfer mechanisms.

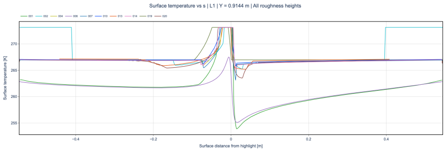
Ice Geometry

- Horn-angle metrics complement the integrated mass quantities by quantifying the spatial distribution of the accreted ice.
- The L1 15-bin ice shapes provide the final geometric comparison with the experimental accretions.

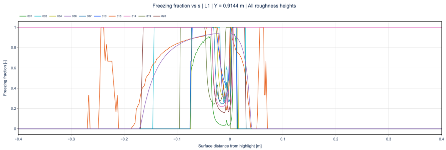
Ice Accretion Takeaway

Aerodynamic, impingement, and convective heat-transfer quantities remain broadly comparable between AE3932 and AE3933, while the icing response begins to distinguish the two conditions. The mass-partition and geometric metrics show how this difference propagates from total ice mass to the final ice shape.

NACA0012 – Surface Temperature and Freezing Fraction – AE3932



Surface Temperature

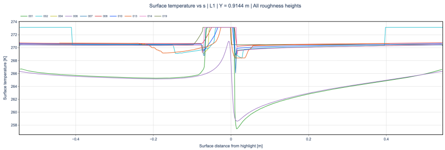


Freezing Fraction

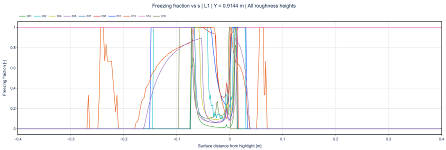
Assessment

- Substantial code-to-code variability is observed in both the surface temperature and freezing-fraction distributions.
- The predicted freezing fraction exhibits different thermodynamic regimes and transition locations across submissions.
- These sharp and spatially shifted transitions make direct pointwise comparison of freezing fraction particularly difficult.

NACA0012 – Surface Temperature and Freezing Fraction – AE3933



Surface Temperature



Freezing Fraction

Assessment

- Similar code-to-code variability is observed for AE3933, particularly in the extent and location of the different freezing regimes.
- Differences in surface temperature are accompanied by substantial differences in the predicted freezing-fraction distributions.
- As for AE3932, the discontinuous nature and spatial displacement of the freezing transitions make a direct pointwise statistical assessment difficult.

NACA0012 – Overall Assessment

Aerodynamics

- Most submissions exhibit limited grid sensitivity in the integrated aerodynamic coefficients, with C_D showing greater sensitivity than C_L and C_M.
- L1 C_p distributions are closely clustered and generally agree well with the experimental measurements.

Droplet Impingement

- Collection-efficiency metrics and integrated water mass exhibit comparatively weak spatial grid sensitivity for most submissions.
- The two icing conditions produce similar water impingement, while reducing the droplet distribution to a single bin introduces a clearer systematic effect.

Convective Heat Transfer

- Reconstructed recovery-temperature distributions are closely clustered across submissions.
- Larger code-to-code differences occur in *HTC* and consequently in the integrated convective heat transfer.

Ice Accretion

- The first systematic distinction between AE3932 and AE3933 emerges in the resulting ice-accretion response.
- Mass-partition and geometric metrics provide additional information beyond total ice mass for interpreting the final ice shapes.